

Dong-Kyum Kim

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Research Interests

- **Interdisciplinary physicist** bridging AI, statistical physics, and neuroscience.
- **How AI stores knowledge:** interpretability, model editing, and machine unlearning in large language models.
- **Physics & brains for AI:** deep learning for nonequilibrium statistical physics, and brain-inspired models of memory.

Education

Korea Advanced Institute of Science and Technology (KAIST) 2016–2022
Ph.D. in Physics Advisor: Prof. [Hawoong Jeong](#)
Dissertation: *Nonequilibrium Statistical Physics Study using Deep Learning*

Seoul National University (SNU) 2011–2015
B.S. in Physics, Minor in Computer Science & Engineering

Research Experience

Postdoctoral Researcher, Max Planck Institute for Security and Privacy (MPI-SP) Oct. 2024–Present
• Hosted by Prof. [Meeyoung Cha](#) (Scientific Director)
• Data Science for Humanity Group

Postdoctoral Researcher, Institute for Basic Science (IBS) Mar. 2022–Sep. 2024
• Hosted by Prof. [Meeyoung Cha](#) (Chief Investigator)
• Data Science Group, Center for Mathematical and Computational Science

Data Science Intern, Samsung Electronics Sep.–Dec. 2017
• Improved anomaly image classification via distributed multi-GPU training (Keras & Spark), with [Daniel Kim](#) (Senior Data Scientist)
• Built a distributed image search framework using Elasticsearch

Publications

[†] denotes equal contribution.

- [1] Jea Kwon, **Dong-Kyum Kim**, Jiwon Kim, Yonghyun Kim, Woong Kook, and Meeyoung Cha. “[AI Engram: In Search of Memory Traces in Artificial Intelligence](#).” *ICML 2026 (Oral)*.
- [2] Minsung Kim, **Dong-Kyum Kim**, Jea Kwon, Nakyeong Yang, Kyomin Jung, and Meeyoung Cha. “[How Training Data Shapes the Use of Parametric and In-Context Knowledge in Language Models](#).” *ACL 2026*.
- [3] **Dong-Kyum Kim**, Minsung Kim, Jea Kwon, Nakyeong Yang, and Meeyoung Cha. “[Bilinear representation mitigates reversal curse and enables consistent model editing](#).” *ICLR 2026*.
- [4] Nakyeong Yang, **Dong-Kyum Kim**, Jea Kwon, Minsung Kim, Kyomin Jung, and Meeyoung Cha. “[Erase or Hide? Suppressing Spurious Unlearning Neurons for Robust Unlearning](#).” *ICLR 2026*.
- [5] Jea Kwon, Jiwon Kim, **Dong-Kyum Kim**, and Meeyoung Cha. “[Moir: Let the Model Direct Its Own Story for Robust Cross-Domain Knowledge Editing](#).” *ICML 2026 Workshop on Mechanistic Interpretability*.
- [6] Minsung Kim, Jea Kwon, **Dong-Kyum Kim**, and Meeyoung Cha. “[In Search of the Engram in LLMs](#).” *ICLR 2025 Blogpost*.
- [7] Gwangsu Kim, **Dong-Kyum Kim**, and Hawoong Jeong. “[Spontaneous emergence of rudimentary music detectors in deep neural networks](#).” *Nature Communications* 2024.

- [8] Jea Kwon, Sunpil Kim, **Dong-Kyum Kim**, Jinhyeong Joo, SoHyung Kim, Meeyoung Cha, and C. Justin Lee. “[SUBTLE: An unsupervised platform with temporal link embedding that maps animal behavior.](#)” *IJCV* (2024).
- [9] **Dong-Kyum Kim**[†], Jea Kwon[†], Meeyoung Cha, and C. Justin Lee. “[Transformer as a hippocampal memory consolidation model based on NMDAR-inspired nonlinearity.](#)” *NeurIPS* 2023.
- [10] Sangyun Lee, **Dong-Kyum Kim**, Jong-Min Park, Won Kyu Kim, Hyunggyu Park, and Jae Sung Lee. “[Multidimensional entropic bound.](#)” *Phys. Rev. Research* **5**, 013194 (2023).
- [11] **Dong-Kyum Kim**[†], Sangyun Lee[†], and Hawoong Jeong. “[Estimating entropy production with odd-parity state variables via ML.](#)” *Phys. Rev. Research* **4**, 023051 (2022).
- [12] Youngkyoung Bae, **Dong-Kyum Kim**, and Hawoong Jeong. “[Inferring dissipation maps from videos using CNNs.](#)” *Phys. Rev. Research* **4**, 033094 (2022).
- [13] **Dong-Kyum Kim** and Hawoong Jeong. “[Deep reinforcement learning for feedback control in a collective flashing ratchet.](#)” *Phys. Rev. Research* **3**, L022002 (2021).
- [14] **Dong-Kyum Kim**[†], Youngkyoung Bae[†], Sangyun Lee, and Hawoong Jeong. “[Learning Entropy Production via Neural Networks.](#)” *Phys. Rev. Lett.* **125**, 140604 (2020).
- [15] **Dong-Kyum Kim**[†], Byunghwee Lee[†], Daniel Kim, and Hawoong Jeong. “[Multi-label classification of historical documents using hierarchical attention.](#)” *J. Korean Phys. Soc.* **76**, 368 (2020).

Technical Skills

- **Daily driver:** Currently running multiple Claude Code sessions (Max plan, \$200/mo) via tmux as my primary research workflow.
- **ML/Interp:** PyTorch, JAX, TransformerLens, & nnsight.
- **Infrastructure:** Distributed training training & inference (Slurm system with hundreds of H100s), large-scale data pipelines

Academic Service

Reviewer: ICML 2026 Workshop Mechanistic Interpretability, ICML 2026 (Received Gold Reviewer Award), ICLR 2026, NeurIPS 2025, ICML 2025 Workshop World Models, ICLR 2025 Workshop Re-Align, NeurIPS 2024 Workshop Behavioral ML, ICML 2024 Workshop LLMs and Cognition.

Awards

Pre-doctoral Fellow of Physics, KAIST

2021

References

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